

# Vaidehi Patil

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## RESEARCH INTERESTS

Language Models, AI safety, Privacy, Security, Multimodality, Multi-Agent privacy, Interpretability, Model editing

## EDUCATION

### University of North Carolina at Chapel Hill (UNC Chapel Hill)

Aug 2022 - May 2027 (Expected)

PhD student in Computer Science

Graduate Research Assistant, Advisor: [Prof. Mohit Bansal](#)

### Indian Institute of Technology Bombay (IIT Bombay) (CPI: 9.52/10)

2017 - 2022

Interdisciplinary Dual Degree: B. Tech. in Electrical Engineering, M. Tech. in AI and Data Science

Minor in Computer Science and Engineering

#### 🏆 Undergraduate Research Award, Best IDDDP Dissertation Award

Thesis: Multilingual Representations for Closely Related Languages

Advisors: [Prof. Sunita Sarawagi](#), IIT Bombay; [Dr. Partha Talukar](#), Google Research India

## WORK EXPERIENCE

### Apple, Cupertino, USA

May - Aug, 2024

ML Research Scientist Intern

Advisors: [Siddharth Patwardhan](#)

Improving faithfulness in RAG by reducing conflict between parametric and contextual knowledge

### Amazon Alexa team, Seattle, USA

May - October, 2023

Applied Scientist Intern

Advisors: [Markus Dreyer](#), [Mengwen Liu](#) & [Leonardo Ribeiro](#)

Self-refinement of multimodal LLMs to create a dataset for multimodal summarization

### Adobe Research India

May - August, 2021

Research Intern (🏆Pre-placement offer as Software Development Engineer)

Advisors: [Dr. Vishwa Vinay](#) & [Dr. Kuldeep Kulkarni](#)

Worked on scene expansion via scene graph using graph generative models

### Adobe Research India

May - August, 2020

Research Intern (🏆Re-internship offer as research intern for Summer 2021.)

Advisor: [Natwar Modani](#)

Introduced a novel problem of detecting versions of documents and created a dataset for it

### AWL Japan

February - April, 2022

Computer Vision R&D Intern

## PUBLICATIONS

\* - equal contribution

1. **UPCORE: Utility-Preserving Coreset Design for Balanced Unlearning.**  
**Vaidehi Patil**, Elias Stengel-Eskin, Mohit Bansal  
Under Review at ICML 2025
9. **Safree: Training-Free and Adaptive Guard for Safe Text-to-Image And Video Generation .**  
Jaehong Yoon\*, Shoubin Yu\*, **Vaidehi Patil**, Huaxiu Yao, Mohit Bansal  
ICLR 2025
8. **Unlearning Sensitive Information in Multimodal LLMs: Benchmark and Attack-Defense Evaluation.**  
**Vaidehi Patil**, Yi-Lin Sung, Peter Hase, Jie Peng, Tianlong Chen, Mohit Bansal  
TMLR 2024

7. **RefineSumm: Self-Refining MLLM for Generating a Multimodal Summarization Dataset**.  
**Vaidehi Patil**, Leonardo F. R. Ribeiro, Mengwen Liu, Mohit Bansal and Markus Dreyer.  
*ACL Main Conference 2024* (🏆 Amazon Conference Travel Grant)
6. **Can Sensitive Information Be Deleted From LLMs? Objectives for Defending Against Extraction Attacks**.  
**Vaidehi Patil\***, Peter Hase\* and Mohit Bansal  
*ICLR 2024 as Spotlight*
5. **Debiasing Multimodal Models via Causal Information Minimization**.  
**Vaidehi Patil**, Adyasha Maharana and Mohit Bansal  
*EMNLP 2023 Findings*  
*NeurIPS 2023 workshop on Causal Representation Learning*
4. **GEMS: Scene Expansion using Generative Models of Graphs**.  
Rishi Agarwal\*, Tirupati Saketh Chandra\*, **Vaidehi Patil\***, Aniruddha Mahapatra\*, Kuldeep Kulkarni and Vishwa Vinay.  
*WACV 2023*
3. **Overlap-based Vocabulary Generation Improves Cross-lingual Transfer Among Related Languages**.  
[Oral Presentation]  
**Vaidehi Patil**, Partha Talukdar and Sunita Sarawagi.  
*ACL Main Conference 2022* (🏆 Google Conference Travel Grant)
2. **Detecting Document Versions and Their Ordering In a Collection**.  
Natwar Modani, Anurag Maurya, Gaurav Verma, Inderjeet Nair, **Vaidehi Patil**, Anirudh Kanfode.  
*International Conference on Web Information Systems Engineering (WISE) 2021*  
🏆 **Best Paper Runner-Up Award**.
1. **Exploiting Language Relatedness for Low Web-Resource Language Model Adaptation: An Indic Languages Study**.  
Yash Khemchandani\*, Sarvesh Mehtani\*, **Vaidehi Patil**, Abhijeet Awasthi, Partha Talukdar and Sunita Sarawagi.  
*ACL Main Conference 2021*

## Patents

2. **Expanding a scene graph using proposals from a generative model of scene graphs**.  
Vishwa Vinay, Tirupati Saketh Chandra, Rishi Agarwal, Kuldeep Kulkarni, Hiransh Gupta, Aniruddha Mahapatra, **Vaidehi Patil**  
*US patent application | Adobe Inc.*
1. **Systems for generating indications of relationships between electronic documents**.  
Natwar Modani, **Vaidehi Patil**, Inderjeet Nair, Gaurav Verma, Anurag Maurya, Anirudh Kanfode.  
*US patent application | Adobe Inc.*

## PROFESSIONAL RESPONSIBILITIES

- *Lead Workshop Organizer* - **Machine Unlearning for Generative AI**, ICML 2025
- *Reviewer* - TMLR 2025, ICLR 2025, ACL 2025, NAACL 2025, TMLR 2024, ACL ARR 2024, CVPR 2022
- *Undergraduate TA* - Center for Machine Intelligence and Data Science, IIT Bombay

## MAJOR ACADEMIC ACHIEVEMENTS

- Awarded **Carolina Computing Fellowship** by the Computer Science Dept, UNC Chapel Hill [’22]
- **2nd rank** in AI and Data Science batch of Centre for Machine Intelligence and Data Science [’22]
- Granted Advanced Performers grade (awarded to **top 1%**) grade for excellent performance [’18]
- Awarded an option of branch change due to exceptional academic performance in first year [’18]
- Secured All India Rank in **top 0.25%** in JEE Mains 2017 and **top 2%** in JEE Advanced 2017 [’17]

## INVITED TALKS

- **Deep Learning: Classics and Trends**  
Can Sensitive Information Be Deleted from LLMs? [Sep ’24]
- **Pluotos**  
UPCORE: Utility-Preserving Coreset Selection for Balanced Unlearning [Apr ’25]